

Oro Verde Tree Planting

Oro Verde, a non-profit environmental organization, has set up a small tree-planting project on a donated parcel of 150,000 square feet of land, in a remote and deforested area. Oro Verde estimated that the water supply in the region is also somewhat limited, with useable reserves not exceeding 50,000 gallons. The project is designed to help large enterprises reduce their carbon footprint, at a minimal cost. In particular, Oro Verde has committed to deliver, precisely 10 years into the future, a total annual sequestration of at least 1,800 pounds of carbon. To deliver on the promise, Oro Verde is considering planting three types of trees: maple, elm and spruce. The trees have very different sequestration rates, require a different amount of water in the early stages of development, and grow to very different widths when they mature. They also have different 10-year survival rates, and the required seedlings have fairly different prices. Oro Verde estimated that a Maple tree, which would deliver 4.3 pounds of carbon sequestration annually, would require roughly 76 gallons of water and 130 square feet of land. Also, the 10-year survival rate for a maple tree is roughly 70%, and the cost of a seedling is \$5.40. An Elm would deliver an annual sequestration rate of 3.7 pounds, requiring 48 gallons of water and 3,600 square feet; the survival rate would be a bit lower, at 60%, but the price would also be commensurate, at \$3.20 per seedling. Finally, a mature spruce would deliver a sequestration rate of 2.8 pounds, requiring 34 gallons of water and 400 square feet of land; the survival rate would be quite high, 80%, but the cost would also be higher, at \$4.50 per seedling. Oro Verde purchases the seedlings from small communities, which have variable amounts available: currently, only 1,000 seedlings of maple, 500 of elm, and 1,500 of spruce can be purchased. In an effort to stimulate the aging population of elms, Oro Verde has also made a special commitment to plant at least 15 elm seedlings.

Oro Verde is seeking your advice concerning the best course of action...